Element I: Testing, Data Collection, and Analysis

I1. Importance of Tests:

We chose to test the longevity of our design because it is essential to the function of a rain barrel; if it only lasts a few years then it isn't a good investment. Also on the project goals, we tested the following aspects related to longevity: durability, ease of use, and water capacity. Durability, because it is slowly eroded or the faucet falls off then you need to fix stuff and that's annoying. Ease of use, because if it's annoying to use, no ones going to want to use the rain barrel. And lastly water capacity, this is most important because if the rain barrel can only hold 20 gallons and the whole garden needs 20 gallons and it hasn't rained in a bit, people will have to go back to the old way of watering (bad).

I2. Comprehension of Testing Procedure:

To test our design, we set up a procedure that includes filling the rain barrel up with water and occasionally having it be used, letting it sit out in the garden for a few weeks to see if it will work properly over time. We gathered data by checking if the rain barrel had any leaks or if water was evaporated out of the barrel and recorded our results using a google doc."

I3. Detail of the Design Analysis:

Our analysis of the test results showed that the barrels worked well and will continue to keep working for at least the next 10 years. The data indicates that the design was successful in meeting the goals because it fit in the required space and had enough water storage, while being easy to use.

I4. Summary of Data Implications:

Based on the data, we plan to stabilize the rain barrels with a stable ground underneath it to improve the design."These results suggest that our next step should be to make the ground underneath the rain barrel flat by packing down dirt then putting the barrel on top of bricks for the garden. "